

A Novel Transformer Architecture for Machine Translation

Abstract:

Abstract: This paper describes a novel transformer-based neural network architecture for Natural Language Processing. We examine sequence-to-sequence machine translation and vocabulary tokenization. Our results indicate that using word embeddings and multi-head attention structures improves translation accuracy across English-Spanish benchmarks.

Introduction:

In modern systems, this domain has seen significant growth. Traditional methods often struggle with scalability and performance limitations. We introduce our approach, which seeks to resolve these issues by introducing novel methods. We validate our design through extensive simulations and comparative tests. The following sections will detail our implementation design, configurations, and results.